Maternal Mortality

Graph Out Loud

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Agenda

01
Recap

02
Project Management Overview

03
Methodology & Data Source Description

04
Data Visualizations

05
Challenges & Learnings

06
Conclusion & Key Takeaways

01
Recap

Research Question

In pregnant woman and new mothers in the United States **(P)**, how has the Covid-19 pandemic **(I)**, compared to the pre-pandemic period **(C)**, affect maternal mortality rates **(O)**, particularly in the years following the pandemic?

- What is the association between demographic and socioeconomic factors and the prevalence of maternal mortality?

Understanding Maternal Mortality

- According to WHO, maternal mortality refers to the **death of a woman during pregnancy, childbirth, or within 42 days of termination of the pregnancy,** from any cause related to or aggravated by the pregnancy or its management, **excluding those from accidental or incidental causes.**

- Most of the **complications that occur are preventable** through the health-care solutions (high-quality care).
  - Causes: severe bleeding, infections, high blood pressure (pre-eclampsia, eclampsia)

- Despite spending more than any other country in the world, the **United States has one of the highest** maternal mortality rates.
  - Key indicator of the quality of the healthcare systems, in context of maternal and reproductive health

- Based on the CDC numbers, Maternal mortality rates are expressed as the **maternal mortality ratio (MMR),** number of maternal deaths per 100,000 live births within a given time frame

02
Project Management Overview

Overview

01 ClickUp: Project Manage

- Organize and manage projects 1 & 2
- Assign tasks, track progress and workflows
- Monitor status

02 Google Drive: Document Sharing

- Share documents, files, slide decks
- Collaborate seamlessly with team members
- Edit documents/slide decks simultaneously

03 Zoom/Google Meet: Meetings & Collaboration

- Conduct virtual project meetings (Zoom for in class, Google Meet for outside of class)
- Host discussions & brainstorming sessions
- Prepare for presentations

04 iMessage: Continuous Project Discussions

- Facilitate real-time communication
- Share quick updates & feedback
- Discuss upcoming tasks

03

Methodology & Data Source Description

Methodology

Choosing a Topic

- Brainstorming on topic (maternal health, mental health)
- Initial analysis performed

Search on Datasets

- Efforts made to find reliable data sources/data sets
- Datasets identified: PRAMS, CDC VSRR Provisional Maternal Death Counts and Rates, CDC Pregnancy Mortality Surveillance Systems

Final Decision

- Pros and cons of datasets identified
- Datasets chosen: PRAMS (2018-2022), CDC VSRR Provisional Maternal Death Counts and Rates (2019-2023) CDC Pregnancy Mortality Surveillance Systems (2017-2020)

Data Extractions & Merging

- **PRAMS: merging 2018 -2022 data** into excel with health indicators
- **Renamed variables** (health indicators)
- **Concatenated** some of the data points (health indicators)
- **Created** pivot tables

Pregnancy Risk Assessment Monitoring System (PRAMS), Maternal and Child Health (MCH) Indicators 2022												
Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8	Column9	Column10	Column11	Column12	Column13
Categories	Nutrition					Pre-pregnancy Weight						
Health Indicator	Multivitamin use ≥4 times a week during the month before pregnancy					Underweight (body mass index [BMI] <18.5 kg/m2)					Overweight (BMI 25–29.9	
Site Name	N (Denominator) Unweighted Sample Size	n (Numerator) Unweighted Frequency	Weighted %	Lower 95% Confidence Interval	Upper 95% Confidence Interval	N (Denominator) Unweighted Sample Size	n (Numerator) Unweighted Frequency	Weighted %	Lower 95% Confidence Interval	Upper 95% Confidence Interval	N (Denominator) Unweighted Sample Size	n (Numerator) Unweighted Frequency
Sites aggregated*	31192	13996	46.2	45.4	47.1	29761	1017	3.1	2.8	3.4	29761	7840
Alabama	765	289	36.9	33.6	40.4	721	18	2.5	1.6	3.9	721	172
Arizona	842	337	38.7	35.5	42.0	806	31	3.9	2.8	5.6	806	211
Colorado	1221	613	50.9	47.7	54.1	1187	30	2.4	1.6	3.6	1187	315
Connecticut	1298	596	53.1	49.7	56.5	1216	40	2.4	1.6	3.5	1216	331
Delaware	870	399	45.8	42.2	49.4	817	26	2.9	1.9	4.4	817	216
District of Columbia	458	229	48.0	42.7	53.4	433	14	3.1	1.6	5.7	433	114
Hawaii	1332	642	49.9	46.2	53.7	1288	44	2.7	1.7	4.2	1288	316
Kansas	1046	489	46.1	42.3	49.8	1000	24	2.0	1.2	3.3	1000	271
Kentucky	758	248	37.5	33.2	42.0	712	16	2.6	1.4	4.7	712	207
Maine	694	392	53.8	49.3	58.2	675	22	3.6	2.1	6.0	675	175
Massachusetts	1515	749	54.1	50.8	57.4	1438	46	2.5	1.7	3.6	1438	388
Michigan	1348	572	45.7	42.5	48.9	1314	50	3.2	2.3	4.6	1314	320
Missouri	838	363	41.9	38.2	45.8	825	30	3.3	2.2	4.9	825	211
Montana	975	437	49.7	46.0	53.3	960	32	2.6	1.7	4.0	960	244
Nebraska	1185	514	49.2	45.3	53.1	1109	28	1.4	0.9	2.2	1109	322
New Jersey	919	485	52.9	49.6	56.2	849	24	2.6	1.7	4.0	849	250
New Mexico	991	460	43.8	40.8	46.9	959	35	3.9	2.8	5.5	959	246
New York City	1310	582	42.9	39.9	46.0	1207	81	6.5	5.1	8.4	1207	302
New York State	855	436	47.7	43.5	52.1	809	26	2.9	1.7	4.9	809	193
North Dakota	503	214	48.4	43.2	53.7	486	14	2.6	1.4	4.9	486	130
Northern Mariana Islands	215	57	25.5	21.4	29.9	198	5	2.5	1.3	4.7	198	50
Oklahoma	1472	514	39.8	36.1	43.7	1425	47	4.1	2.7	6.3	1425	359
Pennsylvania	1276	575	47.8	43.9	51.7	1237	37	2.8	1.8	4.3	1237	354
Puerto Rico	969	286	29.4	26.2	32.8	952	73	7.4	5.7	9.5	952	273
Rhode Island	776	402	50.6	46.5	54.7	724	20	3.0	1.8	4.8	724	215
South Dakota	1247	480	46.8	43.2	50.4	1138	32	2.3	1.4	3.7	1138	286
Utah	1325	643	50.9	47.5	54.2	1298	54	3.2	2.3	4.5	1298	344
Vermont	840	480	53.6	50.1	57.0	802	23	2.5	1.7	3.9	802	207
Virginia	972	467	43.5	38.5	48.7	918	34	3.3	1.9	5.9	918	227
Washington	1107	523	49.8	46.2	53.5	1076	24	2.0	1.2	3.3	1076	273
Wisconsin	825	328	49.1	44.8	53.5	750	21	2.0	1.1	3.7	750	209
Wyoming	445	195	46.6	41.0	52.4	432	16	3.6	2.0	6.5	432	109

Footnotes 2016 2017 2018 2019 2020 2021 2022

- **MMR in the US 1987-2020:** Checked for missing values: No missing values found.
- **Verified data types**: Correct types for numeric and categorical data.
- **Outlier detection**: Identified high mortality rates for further analysis.

```
print("\nMissing Values:")
print(missing_values)

print("\nSummary Statistics:")
print(summary_statistics)

print("\nOutlier Check (Negative Values):")
print(outlier_check)
```

```
Data Info:
None

Missing Values:
Data As Of                 0
Jurisdiction               0
Group                      0
Subgroup                   0
Year of Death              0
Month of Death             0
Time Period                0
Month Ending Date          0
Maternal Deaths            0
Live Births                0
Maternal Mortality Rate    0
dtype: int64
```

Data Wrangling

- Excel
- Python

Data Visualization

- Excel
- Python
- Tableau

04

Tableau: Data Visualization

05

Challenges & Learnings

Challenges

1. Data Availability and Quality
    a. Difficulty in accessing comprehensive datasets for recent years.
    b. Variability in data formats across sources required significant organizing.
2. Data Integration
    a. Merging datasets like PRAMS, CDC VSRR, and Pregnancy Mortality Surveillance Systems was complex.
    b. Addressing inconsistencies in variable definitions and data points.
3. Visualization and Insights
    a. Our team had limited experience with Tableau, which added to the learning curve in creating effective and interactive dashboards.
    b. Limited availability of real-time data for post-pandemic trends.
4. Socioeconomic and Demographic Variations
    a. Challenges in interpreting the interplay between socioeconomic factors and maternal mortality rates.
    b. Balancing statistical analysis with the narrative to highlight disparities.

Learnings

1. Importance of Data Preparation
    a. Thorough data preparation is essential for generating meaningful insights.
2. Collaboration
    a. Team brainstorming led to deeper insights and creative solutions.
3. Learning New Tools
    a. Gaining hands-on experience with Tableau enhanced our data visualization skills.
4. Context Matters
    a. Socioeconomic and demographic factors provide critical context for understanding maternal health challenges.

06

Conclusion & Key Takeaways

Conclusion

1. **Maternal Mortality Trends**
- Maternal mortality rates increased during and post-COVID-19 (until 2023) compared to pre-pandemic years.
- Clear disparities are evident across demographic and socioeconomic groups.

2. **Healthcare System Insights**
- Despite high spending, gaps in access to quality maternal care (insurance) persist in the U.S.
- Preventable causes remain a significant contributor to maternal deaths.

3. **Need for Targeted Interventions**
- Policies must address socioeconomic inequalities and improve access to care.
- Investments in maternal health infrastructure are critical.

Key Takeaways

1. **Data-Driven Decisions**
   - Reliable and comprehensive data is essential for actionable insights into maternal health.

2. **Equity Focus**
   - Efforts must prioritize underserved communities to reduce disparities in maternal outcomes.

3. **Call to Action**
   - Strengthen maternal healthcare policies and ensure consistent, high-quality care nationwide.
   - Enhance data collection and real-time analysis for proactive decision-making.